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SHORT COMMUNICATION

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Neuroanatomy in virtual reality: Development and pedagogical evaluation of photogrammetry-based 3D brain models

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Abstract

Anatomy studies are an essential part of medical training. The study of neuroanatomy in particular presents students with a unique challenge of three-dimensional spatial understanding. Virtual Reality (VR) has been suggested to address this challenge, yet the majority of previous reports have implemented computer-generated or imaging-based models rather than models of real brain specimens. Using photogrammetry of real human bodies and advanced editing software, we developed 3D models of a real human brain at different stages of dissection. Models were placed in a custom-built virtual laboratory, where students can walk around freely, explore, and manipulate (i.e., lift the models, rotate them for different viewpoints, etc.). Sixty participants were randomly assigned to one of three learning groups: VR, 3D printed models or read-only, and given 1 h to study the white matter tracts of the cerebrum, followed by theoretical and practical exams and a learning experience questionnaire. We show that following self-guided learning in virtual reality, students demonstrate a gain in spatial understanding and an increased satisfaction with the learning experience, compared with traditional learning approaches. We conclude that the models and virtual lab described in this work may enhance learning experience and improve learning outcomes.

KEYWORDS
3D models, medical education, neuroanatomy, photogrammetry, virtual reality

INTRODUCTION

Anatomy studies are an essential part of medical training and form the basis for further education, as they provide an in-depth knowledge of the human body. Therefore, learning anatomy constitutes a significant portion of the curricular time in the first years of preclinical medical training. The study of anatomy presents students with a number of unique challenges, including learning a new language (Latin terms and names of anatomical structures), identification of structures in the body and prosections, complex spatial understanding, and the ability to integrate between a large number of newly learned structures and their relative positions (Pandey & Zimitat, 2007).

Typically, students learn anatomy and prepare for dissections using traditional learning tools such as medical textbooks and lecture presentations. However, understanding the correspondence between

Nadav Aridan and Michal Bernstein-Eliav contributed equally to this study.

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two-dimensional (2D) images, as available in traditional tools, and the three-dimensional (3D) reality of the human body is a challenging mental task. This is particularly challenging in studying neuroanatomy, as students find it difficult to understand the 3D spatial relations between the different brain structures (Javaid et al., 2018). Various technological innovations have been suggested to address this challenge, offering 3D visualization aids that are being incorporated at early stages of modern medical studies. Specifically, Virtual Reality (VR) has gained vast interest as a novel approach to facilitate an immersive, interactive, and efficient learning experience in the study of human anatomy in general and in particular, of neuroanatomy (Duarte et al., 2020; Zhao et al., 2020; Moro et al., 2021; Sinha et al., 2022).

Virtual reality was first introduced in the 1960's and is defined as a computer-generated simulation of a 3D environment that can be interacted with in a seemingly physical manner (Freina & Ott, 2015). The user is fully immersed in this artificial environment, which is experienced through sensory stimuli (sight, hearing, and motion) that mimic the properties of the real world through constantly updating, high-resolution head-mounted displays, stereo headphones, and motion-tracking systems (Moro et al., 2017). In anatomy education, VR has been reported to increase learners' motivation and engagement (Moro et al., 2017, 2021; Duarte et al., 2020; Kurul et al., 2020; Zhao et al., 2020) and to promote effective learning, especially for students with relatively low visuospatial abilities (Jang et al., 2017; Kurul et al., 2020; Moro et al., 2021).

The use of VR in anatomy has been accelerated over the past decade as part of world-wide educational trends of self-guided and problem-based learning (Nyemb, 2017). It has been reported to advance learning of the forearm musculoskeletal anatomy (Codd & Choudhury, 2011), the cardiac system (Maresky et al., 2019), the anatomy of the female pelvic floor (Ellington et al., 2019), and the middle ear structures (Ng et al., 2015), among many others.

The spatial complexity of the brain structures presents medical and neuroscience students with considerable challenges in learning neuroanatomy (Sotgiu et al., 2020). About a quarter of previous studies on the benefit of VR for anatomy education has specifically focused on neuroanatomy (e.g., two out of eight studies reviewed in Moro et al. (2021); 4 out of 15 reviewed in Zhao et al. (2020)). While most of these studies have reported positive experience of VR as a learning tool, they did not find significant differences in knowledge compared to traditional learning tools. Importantly, studies that used VR head-mounted displays (HDMs) have employed brain models that were constructed using 3D modeling software or imaging-based volumetric reconstructions, rather than models derived from real brain specimens (Drapkin et al., 2015; Kockro et al., 2015; Stepan et al., 2017; Ekstrand et al., 2018; Obrero-Gaitán et al., 2021; van Deursen et al., 2021). In one previous study where real brain dissections were used for models development, these models were presented on a screen using stereoscopic anaglyph glasses rather than an immersive VR headset (De Faria et al., 2016).

Photogrammetry is a well-established method that allows the construction of highly detailed 3D models of physical objects and environments from multiple-angle high-resolution photographs. It has been demonstrated for acquisition of metric information and 3D visualization of the human brain in medical and educational applications (De Faria et al., 2016; Nocerino et al., 2017; Rubio et al., 2019; Gurses et al., 2021, 2022; Xu et al., 2022). However, to the best of our knowledge, no study has directly tested the contribution of using such realistic brain models within VR to neuroanatomy learning. Thus, it remains an open question, what would be the benefits of learning neuroanatomy by self-guided, free interaction with 3D models constructed from real brain specimens and embedded in a fully immersive virtual-reality environment.

Here, we used a unique approach of combining photogrammetry and advanced editing software to create highly detailed 3D models of a real human brain at different stages of dissection. The models were placed in a custom-built virtual laboratory where participants could walk around to freely explore and manipulate (i.e., lift the models, rotate them for different viewpoints, etc.). Sixty participants without prior knowledge of anatomy or experience in VR were randomly assigned to either a VR group (Read + VR), a Physical models group (Read + 3D printed models) or a Read-only group (Read-only). All participants were given 1 h to learn the white matter tracts of the cerebrum. To evaluate the contribution of VR to neuroanatomy learning, participants completed theoretical and practical exams and a learning experience questionnaire.

MATERIALS AND METHODS

Participants

Sixty healthy volunteers (mean age 23.6, SD 3.16; 24 females and 36 males) participated in this study. All participants were Tel Aviv University students, recruited through advertisements on campus and social media. Participants had no background in anatomy or neuroscience, and no prior experience with VR. The study was approved by the Tel Aviv University Ethics Committee and all participants signed an informed consent.

Donors

Models were created from real human bodies with full and informed consent from the donors. When planning and conducting this study we strictly followed the IFAA ethical guidelines for the acquisition and usage of human tissue images in anatomy education and research (https://ifaa.net/committees/ethics-and-medical-humanities-ficem/recommendations-for-good-practice-around-human-tissue-image-acquisition-and-use-in-anatomy-education-and-research/). Specifically, all dissections of deceased individuals remained confined to a protected non-public space, and images derived from human tissues were treated respectfully, with their usage restricted to this defined research purpose only. The representations of physical specimens that we developed here, both 2D images and 3D models, are completely anonymous and for non-commercial use only.

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Our images and models can in no way lead to the identification of individual donors and do not include any identifying marks, accession numbers, or other features that may potentially lead to identification. All digital images of human tissues acquired for this study are stored on secure, password protected institutional server that is only accessible to designated faculty, staff, and students. Students using the images and models are strictly instructed not to take photos, copies, or screenshots. Images and models are available to the scientific community via an institutional anatomy designated website only.

Experimental procedure

Two participants were invited to each experimental session. Participants were randomly assigned to either a VR or a Read-only condition. First, both participants were given 30 min to study the white matter structures of the brain by reading an introduction to anatomy and a dissection manual of the white matter tracts, written by the Tel Aviv University Sackler School of Medicine Neuroanatomy staff (link). The participant assigned to the read-only condition then had additional 30 min to study by reading, while the participant assigned to the VR condition had 30 min to study 3D models of the white matter structures of the brain (see *Models Construction* below) using a VR headset. The VR participant was fitted with the VR headset and given a short introduction for using the hand controller (5 min) and was then left to self-study. To account for the 3D content available in the VR condition, additional 20 participants were assigned to a Physical models control group. Participants in this group were first given 30 min to read the introduction to anatomy and the white matter tracts dissection manual, similar to the VR and Read-only participants. Then, they were given additional 30 min for self-study with four 3D printed models depicting an undissected hemisphere, a dissection of the association fibers, a dissection of the projection fibers and a dissection of the corpus callosum (see *Physical models* below).

Following the total of 1 h of studying, all participants took a theoretical knowledge exam (10 questions, see Table S1), a practical exam of identifying white matter tracts in a prosected brain (8 questions, see Table S2) and a learning experience questionnaire (9 questions, see Table S3). The theoretical exam included questions that are purely theoretical (questions 1, 2, 6, 7, 8; referred to as "2D questions") and questions that require an understanding of spatial relations gained through brain dissection (questions 3, 4, 5, 9, 10; referred to as "3D questions"). See Figure 1 for a schematic presentation of the study design.

Statistical analysis

Learning experience ratings

A one-way ANOVA was performed to compare the subjective ratings of learning experience through reading the dissection manual across the three learning groups (Read-only/Physical models/VR). A repeated-measures ANOVA of learning group (Physical models/VR) and learning method (Reading/Spatial aid—whether printed models or VR) was performed to compare learning experience ratings within each spatial learning group.

Test scores

One-way ANOVAs were performed to compare the performance in the theoretical and practical tests between learning groups. Performance in the theoretical test was also examined at the level of single test items with statistical significance threshold Bonferroni corrected for 10 comparisons. A repeated-measures ANOVA of group (Read-only/Spatial aids) and question-type (2D/3D) was used to test our hypothesis that learning with a spatial aid (VR or printed models) may enhance performance in the 3D more than the 2D questions.

Models construction

The construction of the models included the following stages: brain dissection, photogrammetry, and model editing, as detailed below. For the current study, only the white matter models were employed.

Brain dissection

Ten models of the brain at different levels of dissection were created to demonstrate the main structures (Figure 2A; models are available

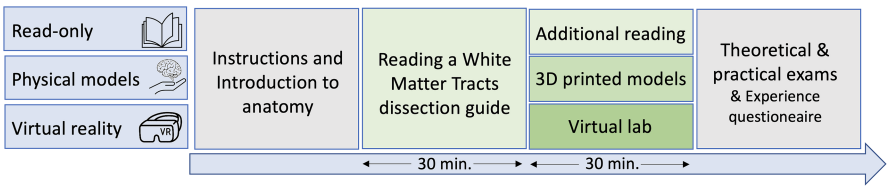


FIGURE 1 Study design. Participants in both the Read-only, the Physical models, and the Virtual reality groups began with reading a short introduction to anatomy followed by 30 min of reading the white matter tracts dissection guide. Then, the Read-only participants continued to additional 30 min of reading, while the Physical models participants had 30 min with the printed models and the Virtual reality participants had 30 min in the virtual lab. All participants then completed theoretical and practical exams, and an experience questionnaire.

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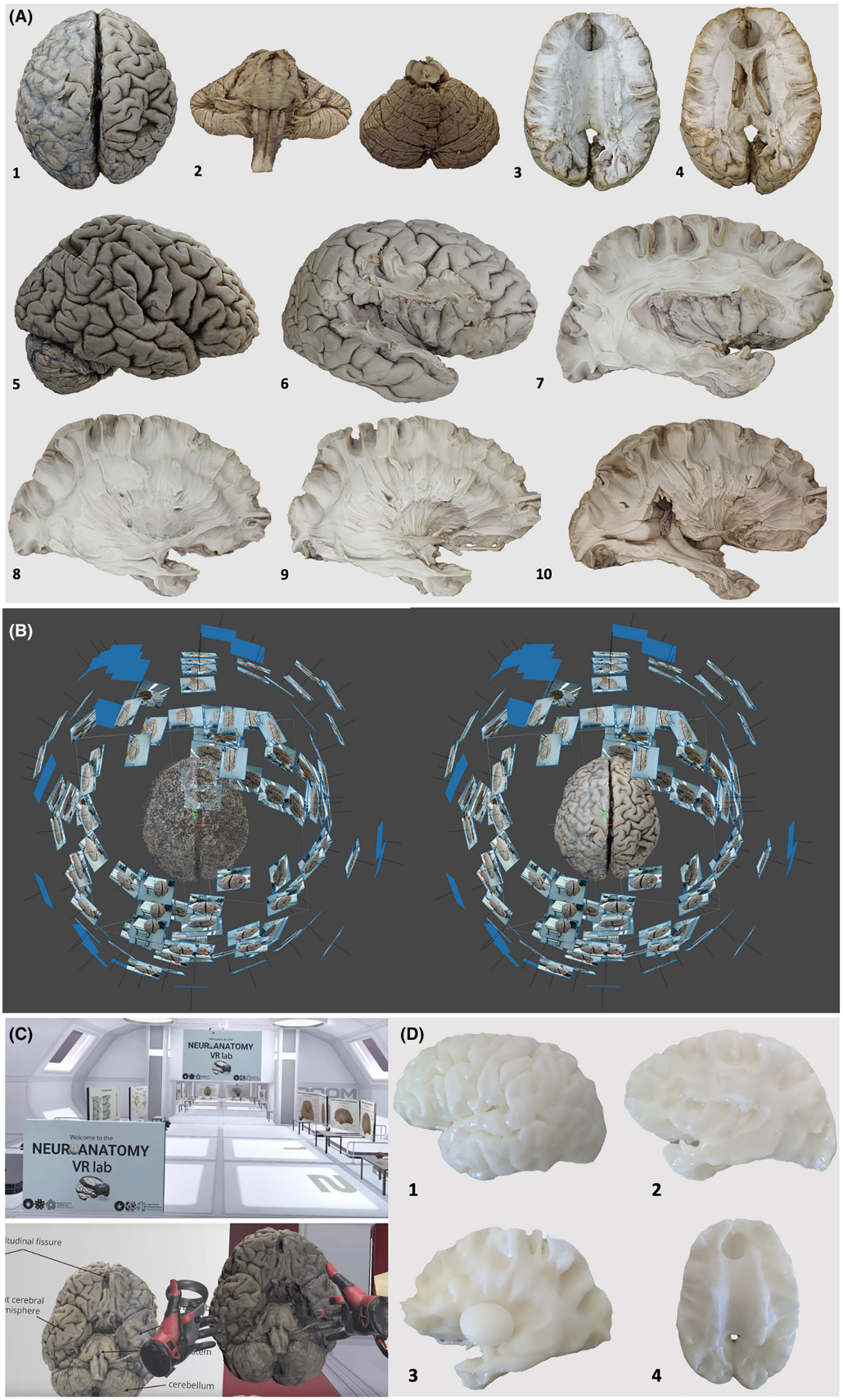


FIGURE 2 Brain models and the virtual lab. (A) Ten 3D models of the brain at different stages of dissection were created. (B) Agisoft Metashape was used to compile multiple photographs of each brain specimen, to generate a point model (left), and to produce a mesh model and its texture (right). (C) Models were placed in a virtual lab (top) allowing participants to manipulate and compare them with annotated 2D images of the same brain specimens (bottom). (D) 3D printed models of white matter dissections.

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online at https://www.neuroanatomy.sites.tau.ac.il). A whole brain (Model 1) was cut at the level of the midbrain, and the brainstem and cerebellum (Model 2) were separated from it. Next, in the superior surface, the cortex was removed to reveal the fibers of the corpus callosum (Model 3). We then made a horizontal cut below the level of the central part of the corpus callosum to reveal the anterior and central parts of the lateral ventricles (Model 4). A whole hemisphere (Model 5) was dissected to remove the operculum and reveal the insula (Model 6). We then removed the cortex from the rest of the hemisphere to reveal the association fasciculi (Model 7). Next, we removed the insula and association fasciculi to reveal the corona radiata and the lentiform nucleus (Model 8). The lentiform was then removed to reveal the internal capsule and anterior commissure (Model 9). Finally, we removed the optic radiation and temporal lobe white matter to reveal the occipital and temporal horns of the lateral ventricle (Model 10). In addition to these 10 models, we created models of the skull, a thoracic vertebra, a cervical vertebra, and the vertebral column, as well as coronal sections of the brain at the level of the following structures: optic chiasm; anterior commissure; mammillary bodies; tegmentum of midbrain; lateral geniculate nucleus; splenium of corpus callosum; and occipital horn of lateral ventricle.

Photogrammetry, image capture, and data processing

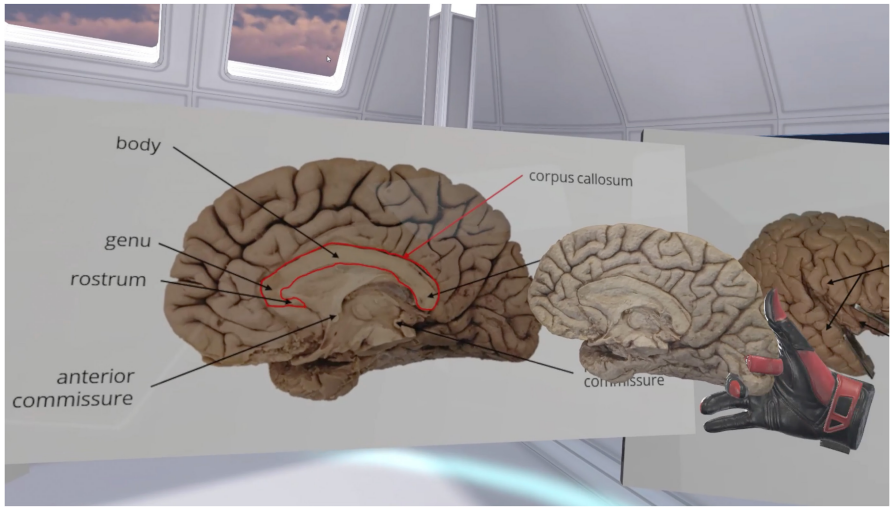
At each stage of the dissection, we photographed the brain from multiple angles (~200 images) using a smartphone camera (Sony IMX586, 48 megapixels). Agisoft Metashape standard edition, version 1.5.0 (http://www.Agisoft.com) was used to compile the photographs, generate a point model, and produce the mesh model and its texture (Figure 2B). Photos were aligned using the high accuracy setting to generate a sparse point cloud. The default selections were unmodified (generic preselection; key point limit = 0, unlimited; tie point limit = 1000; adaptive camera model fitting). The resulting dense point cloud was manually edited to remove any erroneously identified points, not part of the anatomical specimen. The resulting model was exported in OBJ, JPG, and MTL file formats (OBJ file contains the mesh, JPG, and MTL files contain the texture and its properties).

Model editing

Models were imported into Blender software, version 2.91.2, (Blender Institute B.V., Amsterdam, the Netherlands; https://www.blender.org). They were aligned to confirm with standard XYZ axis, scaled to fit within 0.5-m boundary box, smoothed, and cleaned from access material.

Virtual lab setup

VR model viewer software (MindRend Technologies) was used for model display and interaction (http://mindrend.com/). Participants were placed in a virtual room surrounded by tables that matched the dimensions of the physical room (3 × 2 m), allowing for free 6 degrees of freedom (6DOF) locomotion (as opposed to controller-based locomotion; Figure 2C, Video 1). Seven brain models were placed on the surrounding tables: a whole brain model (Figure 2A.1), a model of the corpus callosum (Figure 2A.3) and six models of the right hemisphere at different levels of white matter dissection (Figure 2A.5–9). For each model, a 2D image that included the names of the different structures with corresponding arrows was placed next to it. Participants were fitted with a VR head-mounted display (Meta Quest2) connected wirelessly via "airlink" (TP-Link AX1500 Router) to a PC (CPU: INTEL I7-9700, RAM: 8GB DDR4 2666Mhz,



VIDEO 1 The virtual neuroanatomy lab. Participants were placed in a virtual room surrounded by tables, on six models of the right hemisphere at different levels of white matter dissection were placed. Participants were able to navigate freely in the room and to lift and rotate the models. Next to each model, a 2D image that included the names of the different structures was available.

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GPU: ZOTAC RTX-2070) and used one controller to freely manipulate the models (6DOF).

Physical models

As an additional control for the VR condition, we used physical (resin) models printed in an Epax 3D printer. These models provide 3D information, similar to the VR models, but without the virtual-reality experience. Four models were printed (Figure 2D): (1) an undissected right hemisphere (7×16×9.5 cm); (2) a dissected hemisphere showing the association fiber tracts (5.8×16×10.5 cm); (3) a further dissected hemisphere revealing the projection fibers (5.2×16×10.5 cm); and (4) a whole brain dissected to reveal the corpus callosum (11.5×16×8.2 cm).

RESULTS

Subjective ratings of the learning experience with the dissection manual significantly differed between learning groups ($F_{(2,57)} = 3.69, p < 0.05$). Post-hoc comparisons revealed that ratings were significantly higher in the VR than the Read-only group ($\bar{d} = 0.9$, $t = 2.70$, CI = [0.10 1.70], $p < 0.05$), even though both groups read the exact same manual (Figure 3A). No differences were found between the Physical models and VR or Read-only groups. This suggests that following the experience with VR, the participants retrospectively reappraised the experience of learning through reading. Within the participants that studied using spatial learning aids (VR or Physical models), ratings of the spatial learning experience were significantly higher than learning with the dissection manual ($F_{(1,38)} = 43.97$, $\bar{d} = 0.9$, CI = [0.62 1.17], $p < 0.001$;

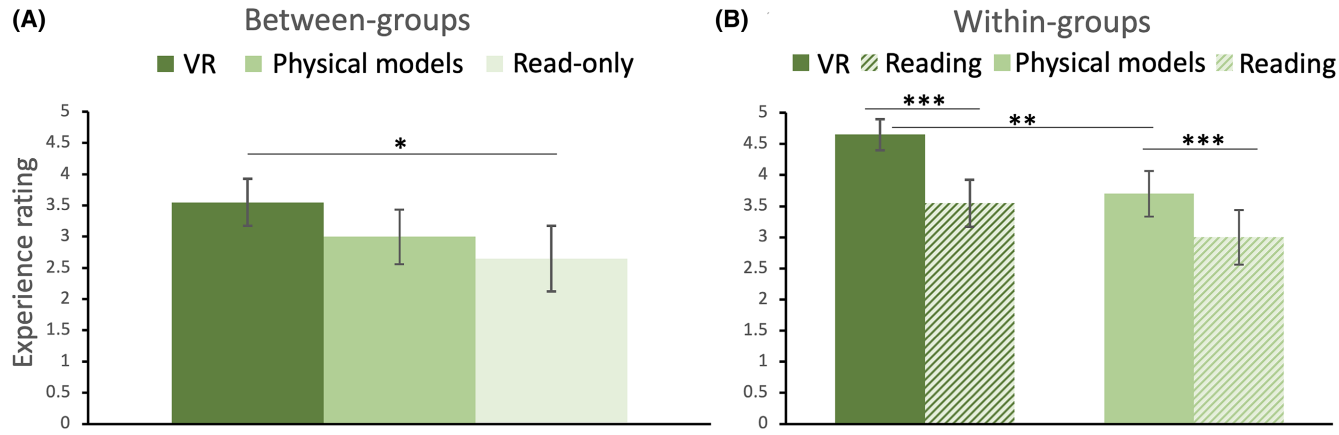


FIGURE 3 Subjective ratings of learning experience. (A) Reading the dissection manual was rated as significantly more interesting, effective, and fun by participants who later experienced with VR than by participants in the Read-only group (*$p < 0.05$). (B) Within the VR and Physical models groups, learning with VR was rated as significantly more interesting, effective, and fun than learning from reading (***$p < 0.001$). Additionally, learning with VR was rated higher than learning with the physical (printed) models (**$p < 0.01$).

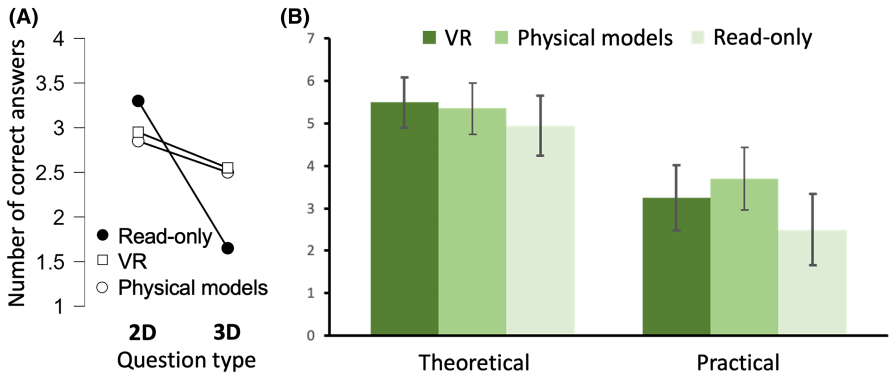


FIGURE 4 Performance in theoretical and practical neuroanatomy exams following learning with VR, with 3D-printed models or by reading only. (A) In the theoretical exam, a significant interaction between group (Spatial aids/Read-only) and question-type (2D/3D) suggests that while there was no difference between groups in purely theoretical questions, the VR and physical models groups were significantly better than the Read-only group in questions that require spatial understanding. (B) Learning with spatial aids (3D-printed physical models or the same models implemented in VR environment) induced a non-significant trend of improvement in both the theoretical and the practical exams relative to learning from a dissection manual with 2D images.

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Figure 3B). Across learning methods, the VR learning experience was rated higher than the Physical models ($F_{(1,38)} = 10.25$, $\bar{d} = 0.75, CI = [0.28\ 1.22], p < 0.01$).

In examining the performance in the theoretical questions, we found an interaction effect of learning method (Spatial aids/Read-only) and question-type (2D/3D). While there was no difference between the Spatial learning (VR or Physical models) and Read-only groups in their performance in the 2D questions, the Spatial learning groups were significantly better than Read-only in the 3D questions ($F_{(1,58)} = 29.94, p < 0.001$; Figure 4A). Additionally, there was a main effect of question-type such that regardless of learning method, the participants answered correctly more 2D questions than 3D questions ($F_{(1,58)} = 10.84, p < 0.01$). There was no difference between groups beyond question-type (i.e., no main effect of learning method, $F_{(1,58)} = 1.38, p = 0.24$; Figure 4B).

Examination of individual items of the theoretical exam revealed that the largest improvement was in questions that require spatial understanding of the relative locations of different brain structures. For example, VR participants performed significantly better than Read-only participants in question #10: *what fiber tract is revealed when removing the lentiform nucleus?* ($p = 0.001$, corrected for multiple comparisons). Similarly, performance in the VR group was higher than Read-only in question #3: *what fiber tract is first exposed when removing the cortex surrounding the insula?* ($p = 0.022$, which did not survive multiple comparison correction). No significant difference between groups was observed in question #1: *which of the following is correct regarding the corpus callosum?* or in question #7: *which two lobes are connected by the arcuate fasciculus?*

Finally, a non-significant trend of improved performance following learning with VR was also detected in the practical neuroanatomy exam (VR vs. Read-only: $\bar{d} = 0.75, p = 0.4$; Physical models vs. Read-only: $\bar{d} = 1.2, p = 0.1$; Figure 4B).

DISCUSSION

This work depicts the development of a novel set of 3D models of brain dissections and their implementation in neuroanatomy teaching in a virtual reality environment. We used photogrammetry and advanced editing software to create 3D models of brain specimens from multiple overlapping photographs. Real human bodies donated to the Tel Aviv University Faculty of Medicine were used to extract the specimens for 3D modeling, offering unique insights into brain anatomy which may be more informative than currently available computer-generated or imaging-derived brain models. Moreover, models were embedded in a virtual reality environment, proving an experience of a virtual dissection laboratory where students can freely explore, manipulate, and interact with the models. In a cohort of 60 participants without prior experience in anatomy or VR, we show that VR enhanced learning experience and improved learning outcomes. Improvement was detected particularly in questions that required visuospatial abilities, that is, understanding the relative spatial locations of different brain structures.

Neuroanatomy has been reported to induce fear and difficulties among medical students, as neuroanatomical concepts are considered more complex and difficult to understand relative to other anatomy topics (Javaid et al., 2018). This phenomenon, often termed "neurophobia" (Abulaban et al., 2015; Sotgiu et al., 2020; Moreno-Zambrano et al., 2021), can be at least partially ameliorated by new pedagogical approaches that are aimed to create interactive and engaging learning environments and increase students' motivation and curiosity. These include problem-based learning (PBL), computer-assisted learning (CAL), and self-guided learning (Mourad et al., 2016). We suggest that self-learning with high-quality models in a virtual laboratory can significantly improve students' subjective learning experience, and more so, their retrospective reappraisal of learning from textbook, leading to increased motivation, and decreased neurophobia.

Real donors dissections were used to create our brain models, as opposed to the vast majority of previous studies which utilized computer-generated or imaging-based models (Drapkin et al., 2015; Kockro et al., 2015; Stepan et al., 2017; Ekstrand et al., 2018; Obrero-Gaitán et al., 2021; van Deursen et al., 2021). In one study, models that were developed based on photographs of real brain specimens were shown to improve students' understanding of the limbic system structures relative to conventional teaching methods (De Faria et al., 2016). However, it is important to note that this study utilized stereoscopic anaglyph glasses and a computer screen to display the models rather than an immersive VR headset. Additionally, the rotation of the models was limited to only two axes (vertical and horizontal), they were presented absent of structures' names or explanatory text, and students' exposure to them was relatively restricted (watching a videoclip of the models, presented by an instructor with a pre-written script, rather than self-guided, free interaction with the models). In the present work we therefore took one step further and examined the pedagogic effectiveness of 3D models accompanied with annotated 2D images and embedded in an interactive VR environment (Video 1, Figure 2).

We additionally included a "Physical models" condition, aiming to separate the spatial information that is available in the VR condition from the VR learning experience. Participants in the Physical models group were given a set of 3D-printed models depicting white matter tracts dissections, similar to those presented in the virtual lab. Results were similar for learning with VR and Physical models in terms of theoretical and practical test scores. Both spatial aids (virtual and physical models) improved students' performance in questions that require spatial understanding. Notably, learning with VR had a significantly higher rating of learning experience compared to the Physical models. These findings suggest that VR offers all of the benefits of physical spatial learning with additional advantages such as enhanced user engagement and the potential to subserve remote learning. Future studies should directly compare the contribution of different physical and virtual models to anatomy education.

Future investigation may also directly compare the pedagogical value of the VR lab with that of real-life dissections. Here, we investigated three learning procedures that are alternatives to real-life

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dissections, as human donor bodies are an expensive resource and are not accessible remotely (e.g., to allow for anatomy studies in times of pandemic, etc.). The three learning methods compared here delineate between two important aspects of anatomy studies: photorealistic content and 3D spatial information. Specifically, the Read-only participants were exposed to high-resolution realistic images of the human brain, but without 3D information; the Physical models participants were exposed to 3D structures without the ecological texture of real brains; and the VR participants were exposed to information which is both photorealistic and 3D. By comparing these three conditions we offer novel insights on the relative contribution of realistic images and 3D spatial content to leaning neuroanatomy. Participants from all learning groups were tested on real brain specimens, to simulate the practical exams that anatomy students undertake in reality. Therefore, learning with real specimens would have biased our results in favor of those participants that learned and were then tested with the same means. Future studies may include a learning group with real brains and test participants using MRI images, for instance, instead of real brains, to avoid this bias.

While we observed a significant improvement in learning *experience* in the VR compared to the Read-only group, the effect of VR on learning *outcomes* was less decisive. The VR group was significantly better than Read-only on theoretical questions that required an understanding of spatial relations that are gained through brain dissection. This is consistent with previous reports supporting an advantage of VR over traditional tools such as lecture slides or textbook images, which may be less effective for the complex spatial understanding required for learning anatomy (Jang et al., 2017; Kurul et al., 2020; Moro et al., 2021). Our findings are also in line with a previous study that examined learning neuroanatomy from MRI images and overlayed 3D composite models. Compared to participants that studied only from 2D images, the participants that studied from 3D models were better at identifying brain structures that are more spatially complex (Drapkin et al., 2015). We additionally found a non-significant trend of improvement in performance in the practical exam. Indeed, as VR is a relatively new pedagogical approach, there are mixed findings regarding its benefits. For example, in a study comparing learning from a plastic model, photographs of the plastic model, and virtual reality, the model group scored significantly better than either the photographs or the VR groups (Khot et al., 2013). Another study that compared learning from gross prosections to learning from 3D models found no difference between the groups (Hopkins et al., 2011). Therefore, while VR may be better than using only a textbook or dissection manuals, it may not be as good as actual dissections (Codd & Choudhury, 2011).

Another consideration regarding the implementation of VR in medical training is its potential adverse effects such as discomfort, headache, fatigue, dizziness, nausea, and disorientation, in addition to ocular symptoms such as blurred vision, difficulty in focusing, and double vision (Moro et al., 2017; Kuehn, 2018). No such side effects were encountered in the current sample, but they may warrant caution in future pedagogic interventions heavily relying on VR. Alternatively, augmented reality (AR; e.g., Küçük et al., 2016) is considered a "safer" approach with less adverse effects. In AR, virtual objects overlap with both physical objects in the real space, so that the user is placed in a real-life environment and can interact with physical and virtual objects at the same time. While currently AR falls behind VR in terms of graphical capabilities, as the technology continues to develop it may be a promising prospect for future research to incorporate the models presented here in AR (Barsom et al., 2016; Herron, 2016; Dhar et al., 2021).

This work presents the process of developing, implementing, and evaluating a novel, interactive tool for neuroanatomy teaching. While our results provide compelling evidence in support of the pedagogic value of this tool, they are based on relatively small samples and may therefore benefit from further validation in larger cohorts. Assessments of students' general cognitive abilities (e.g., working memory) as well as more specific abilities associated with successful anatomy learning (e.g., mental rotation), and pre-learning neuroanatomy tests, may shed light on individual-level pedagogical advantages of the proposed tool. It should also be examined in comparison with various other traditional (e.g., dissections/prosections) or more advanced teaching approaches (e.g., AR). Additionally, we focused here on naïve participants in order to control for prior knowledge in anatomy and experience with dissections. As all of our models, photogrammetry-content and dissection guides are available for the anatomy research and education community, future studies may examine the beneficial effects of VR on medical students training as well. Finally, future studies should address its adaptability to other educational purposes and generalizability to additional topics in neuroanatomy and gross anatomy.

CONCLUSIONS

A major challenge in neuroanatomy is the 3D spatial understanding of the relative positions of brain structures. We propose the implementation of interactive and immersive 3D brain models in a virtual reality environment to address this challenge and enhance students' curiosity and motivation. We present a unique set of realistic models depicting the human brain at different stages of dissection and show that following 30-min, self-guided interaction with these models, students show a gain in spatial understanding and an increased satisfaction with the learning experience, compared with traditional learning approaches. The models developed in this work may not only be incorporated in VR as we show here, but can also be accessed remotely and applied in course lectures or online materials, and may be customized to display various contents.

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